

Maintenance Schedule

Gas engine-generator set

MTU 6R400 GS - B3066Z8

MTU 12V400 GS - B3042Z7

MTU 6R400 GS - B3066Lx

MTU 12V400 GS - B3042Lx

MTU 6R400 GS - B3066LHx

MTU 12V400 GS - B3042LHx

Biogas - interval: 800 - 48,000 operating hours

50Hz / 60Hz

MS50196/01E

SLU Name	Sales Name	Engine model		Application group
GG06R0400A1	MTU 6R400 GS	B3066Z8		3A, continuous operation, unrestricted
GG12R0400A1	MTU 12V400 GS	B3042Z7		3A, continuous operation, unrestricted
GG06R0400A1	MTU 6R400 GS	B3066Lx		3A, continuous operation, unrestricted
GG12R0400A1	MTU 12V400 GS	B3042Lx		3A, continuous operation, unrestricted
GG06R0400A1	MTU 6R400 GS	B3066LHx		3A, continuous operation, unrestricted
GG12R0400A1	MTU 12V400 GS	B3042LHx		3A, continuous operation, unrestricted

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

In the case of emission-certified products, nonobservance of the maintenance specifications can result in a violation of the statutory emission regulations. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission-relevant components may only be maintained, exchanged or repaired if the components used for this purpose are approved by MTU or equivalent components. Emission-relevant components are for instance control units, data records, sensors, injectors, exhaust flaps and all components of exhaust aftertreatment systems.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability. The maintenance intervals and required maintenance scopes are based on operational experience and must therefore be regarded as recommendations. In the event of difficult operational and ambient conditions, additional maintenance tasks and/or changes to the maintenance intervals may be necessary.

Maintenance activities are specified with intervals based on hours in operation and time limits. The limit which occurs first is applicable.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational checks and maintenance work that does not require disassembly of the product.

QL2: Component replacement (corrective, and thus not an element of the maintenance schedule).

QL3: Maintenance work that requires partial disassembly of the product.

QL4: Maintenance work that requires complete disassembly of the product.

Supplementary information on maintenance and preservation:

Change intervals for fluids and lubricants and details on preservation can be obtained in the relevant specifications of the component manufacturers and MTU Fluids and Lubricants Specifications. The latest version is also available at: <http://www.mtu-online.com>, for engine-generator sets at: <http://www.mtuonsiteenergy.com>.

Components which are not listed in this maintenance schedule must be serviced in accordance with Manufacturer's Instructions.

This maintenance schedule does not apply to engines with an EPA/CARB emission certificate.

1.2 Additional notes on the maintenance of gas systems

General information

The maintenance schedule matrix normally finishes with extended component maintenance or with engine overhaul. Following this, maintenance work is to be continued at the intervals indicated.

The numbers stated in the task list serve as a reference for the scope of parts required.



Repairs tasks on the gas system, for which the gas-conducting area is opened, and leak or stress tests must only be carried out by specialized companies. As a rule these are companies that are approved for the gas industry with qualified personnel who have been trained for technical gas applications.



The fluids and lubricants specification, the guidelines for maintenance and fluid and lubricant change as well as a list of recommended fluids and lubricants are contained in the Fluids and Lubricants Specifications of MTU Onsite Energy as well as in the fluids and lubricants specifications of the component manufacturers. They are therefore not included in the maintenance schedule (exception: deviations from the Fluids and Lubricants Specifications). Use only fluids and lubricants which comply with the Fluids and Lubricants Specifications of MTU Onsite Energy or which are approved by the relevant component manufacturer.

The following additional maintenance operations must be performed on direct responsibility of the operator / customer:

- Protect components made of rubber or synthetic material from oil. Never treat them with organic detergents. Wipe with a dry cloth only.
- Gas filter:
The maintenance interval depends on gaseous fuel quality (purity). Fuel filters upstream of the engine must be regularly cleaned and renewed if necessary.
- Battery:
Battery maintenance depends on duty and ambient conditions. The specifications of the battery manufacturer apply.

Items listed in this maintenance schedule but which are not included in the scope of delivery of MTU Onsite Energy may be disregarded.

Generator maintenance

Please refer to the Operating Instructions provided by the generator manufacturer for the intervals for these maintenance tasks.

Generator circuit breaker maintenance

Please refer to the Operating Instructions provided by the manufacturer for the intervals for these maintenance tasks.

Maintenance of non-MTU components

Please refer to the Operating Instructions provided by the relevant component manufacturer for the intervals for these maintenance tasks.

Maintenance tasks

The respective maintenance tasks to be carried out are based on operating hours (see the following pages).

1.3 Checks by the operator



Always observe the safety instructions.



Non-observance of these instructions may lead to damage to the system!



When using the unit supplied in a foreign country or in special regions, MTU Onsite Energy is not responsible for the adherence to the legal or other regulations / standards at the place of use.

Recommended daily checks – System

- Check exhaust gas piping for condensate leaks (visual inspection)
- Check condensate collecting tank downstream of condensate drain (if fitted)
- Carry out visual inspection of engine and plant for leaks / damage (bellows, hoses etc.)
- Check gas supply / gas system for damage (visual inspection)

Only for 60 Hz transmission units

- Oil temperature
- Storage temperature
- Oil pressure
- Oil contamination indicator
- Leaks
- Oil level
- Noise, vibrations

Recommended daily checks – Control system

The following measured values and functions are monitored by the control system to protect the plant. These messages can also be called up via remote access (option) and can be monitored by the customer in the display.

- Electric power
- Lube oil level
- Lube oil pressure
- Coolant temperature
- Coolant pressure
- Oxygen sensor voltage (option)
- Battery voltage
- Smooth running of engine
- Exhaust gas temperature
- Heating water supply and return temperature (option)
- Oil bleed system function (option)

Recommended weekly checks

- Lube oil consumption or content of fresh oil tank
- Engine room temperature
- Check of operating hours
- Check of number of starts
- Check of speed control for stability of regulator and the linkage (if installed)
- General visual inspection of module / engine-generator set

Only for 60 Hz transmission units

- Transmission mounting
- Outer condition of transmission (contamination, oil deposits)
- Visual condition of the transmission oil

Additional checks

The operator's gas installation (upstream of twin solenoid valve) must be inspected for leaks at the following intervals:

- Every year for engines with turbocharger
- Every six months for aspirated engines

The following checks are to be carried out at appropriate intervals, depending on the degree of contamination:

- Check room air fan and fresh air filter for contamination
- Check filter or air-conditioning unit for control cabinet for contamination
- Check recooler for contamination

Electrical systems

Electrical systems and operating equipment must be checked prior to initial commissioning, after repairs, prior to re-commissioning and at regular intervals. The intervals are to be so arranged that defects, which are to be expected, are detected at an early stage.

Here, among others, the following requirements apply:

- Accident prevention regulations BGV A3 "Electrical Systems and Operating Equipment"
- Technical rules for operational safety TRBS 2131 "Electrical Hazards"
- Industrial Safety Directive (BetrSichV)
- DIN VDE 0701/0702 Repairs, modifications and testing of electrical equipment/repeat tests on electrical units
- DIN EN 60204-1 Safety of machinery - Electrical equipment of machines

The operating company has the responsibility to see that these tests are carried out correctly. Electrical systems and fixed electrical operating equipment must be tested at appropriate intervals by a qualified electrician for their correct condition. The test is to include the following:

- Visual inspection and testing of protective equipment
- Testing of protective conductor resistance, insulation resistance, leakage current replacement, protective conductor current and differential current
- Carrying out fault diagnoses
- Preparation of test reports and inspection stickers

The industrial safety regulations, the technical rules for operational safety and the BGV A3 are fundamental with regard to adherence to the protective measures for "Electric Plants and Operating Equipment". The operating company is accordingly obligated to care for the safety of electrical systems and operating equipment.

	Designation
DIN	Deutsches Institut für Normung (Federal German Standards Institute)
EN	Europäische Normung (European Standards)
ISO	International Standards Organization
VDE	Verband der Elektrotechnik, Elektronik und Informationstechnik (Association for electrical engineering, electronics and information technology)
BGV	Berufsgenossenschaftliche Vorschriften (Trade association regulations)
TRBS	Technische Regeln für Betriebssicherheit (Technical rules for operational safety)

Table 2: Designations for standards and regulations



These regulations are contained in the manufacturers' instructions, legal requirements and technical guidelines valid in the individual countries. Since there can be major differences from country to country, it is not possible to provide a universally applicable statement on the rules to be observed within the framework of this maintenance schedule.

The user of the product mentioned herein is obligated to inform him-/herself about the provisions applicable in his/her country. MTU Onsite Energy does not accept any liability whatsoever for improper or unlawful use of the electric plant it has approved.

1.4 Maintenance schedule matrix

0-23,200 operating hours

Task	Limit	Operating hours [h]																														
		Daily	800	1,600	2,400	3,200	4,000	4,800	5,600	6,400	7,200	8,000	8,800	9,600	10,400	11,200	12,000	12,800	13,600	14,400	15,200	16,000	16,800	17,600	18,400	19,200	20,000	20,800	21,600	22,400	23,200	
Engine																																
W0940			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0943			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0926			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0930			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0916			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0917			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0956			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0922			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0937			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0951			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0901			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0939			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0924			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0928			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0942			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0950			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0938			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0954			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0918					X			X			X			X			X			X			X			X			X			
W0921					X			X			X			X			X			X			X			X			X			
W0929					X			X			X			X			X			X			X			X			X			
W0907					X			X			X			X			X			X			X			X			X			
W0915					X			X			X			X			X			X			X			X			X			
W0944					X			X			X			X			X			X			X			X			X			
W0904					X			X			X			X			X			X			X			X			X			
W0927					X			X			X			X			X			X			X			X			X			
W0948					X			X			X			X			X			X			X			X			X			
W0953					X			X			X			X			X			X			X			X			X			
W0903								X						X						X						X						
W0925								X						X						X						X						
W0948								X						X						X						X						
W0905											X									X											X	
W0920											X									X										X		
W0952											X									X											X	
W0911												X										X										
W0946												X										X										
W0908													X														X					
W0949													X														X					
W0923																X																
W0936																				X												
W0912																						X										
W0913																						X										

w = weeks
m = months
a = years

Task	Limit	Operating hours [h]																			
		Daily	800	1,600	2,400	3,200	4,000	4,800	5,600	6,400	7,200	8,000	8,800	9,600	10,400	11,200	12,000	12,800	13,600	14,400	15,200
W0945																					X
W0935																					X
W0900																					
W0906																					
W0931																					
W0914																					
W0933																					
W0955																					
W0934																					
W0967																					
w = weeks m = months a = years																					

24,000-47,200 operating hours

Task	Limit	Operating hours [h]																														
		24,000	24,800	25,600	26,400	27,200	28,000	28,800	29,600	30,400	31,200	32,000	32,800	33,600	34,400	35,200	36,000	36,800	37,600	38,400	39,200	40,000	40,800	41,600	42,400	43,200	44,000	44,800	45,600	46,400	47,200	
Engine																																
W0940		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0943		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0926		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0930		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0916		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0917		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0956		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0922		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0937		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0951		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0901		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0939		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0924		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0928		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0942		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0950		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0938		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0954		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0918		X			X			X			X			X			X			X			X			X			X			
W0921		X			X			X			X			X			X			X			X			X			X			
W0929		X			X			X			X			X			X			X			X			X			X			
W0907		X			X			X			X			X			X			X			X			X			X			
W0915		X			X			X			X			X			X			X			X			X			X			
W0944		X			X			X			X			X			X			X			X			X			X			
W0904		X			X			X			X			X			X			X			X			X			X			
W0927		X			X			X			X			X			X			X			X			X			X			
W0948		X			X			X			X			X			X			X			X			X			X			
W0953		X			X			X			X			X			X			X			X			X			X			
w = weeks m = months a = years																																